

DARSHAN THAKER

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RESEARCH INTERESTS

I develop inference-time algorithms that make generative models controllable and reliable. My research spans two complementary directions:

- **Diffusion model guidance:** Training-free posterior sampling algorithms for solving inverse problems with applications in computer vision and medical imaging.
- **LLM robustness and steering:** Optimization-based hallucination elicitation, jailbreak attacks, and activation-space concept steering.

EDUCATION

University of Pennsylvania

Sep 2020 – Dec 2026 (expected)

Ph.D., Computer Science.

Columbia University

2018 – 2019

M.S., Computer Science.

The University of Texas at Austin

2014 – 2018

B.S., Computer Science and Pure Mathematics.

EXPERIENCE

University of Pennsylvania

Sep 2020 – Present

Ph.D. Researcher. Advised by Prof. René Vidal.

Structured Guidance for Solving Inverse Problems with Diffusion Models (ICCV 2025)

- Introduced Frequency-Guided Posterior Sampling (FGPS), a training-free diffusion guidance algorithm for solving ill-posed image inverse problems.
- Evaluated FGPS across deblurring, dehazing, and related image restoration tasks, demonstrating improved reconstruction quality without retraining the diffusion model.

Defenses for Adversarial Attacks in Computer Vision (ICML 2022; Preprint 2023; ICLR-W 2026)

- Developed sparse coding algorithms that recover the deception mechanism underlying adversarially perturbed images, with theoretical recovery guarantees.
- Co-developed an extension of the framework to defend image classifiers and vision-language models against semantic adversarial attacks.

LLM Hallucination Elicitation and Concept Steering (NeurIPS 2024; NeurIPS 2025; ICML 2026)

- Co-designed an inference-time framework (PaCE) that uses sparse coding to identify and steer concepts in LLM activation spaces.
- Mentored a junior PhD student in developing optimization-based methods (SECA and REALISTA) to elicit LLM hallucinations.

Siemens Healthineers

Summer 2025

Research Intern. Mentor: Dr. Mariappan Nadar.

- Designed and trained a scale-cascaded diffusion model on a multi-GPU cluster and developed a posterior-sampling algorithm to steer it for medical image super-resolution (ISBI 2026, *oral*).

Salesforce Research

Spring 2020

Research Intern. Mentor: Dr. Yu Bai.

- Analyzed theoretical conditions under which deep networks learn transferable features, focusing on lower bounds on generalization error.

Earlier Industry Experience

The Curious AI Company	Research Intern, Industrial Process Control	<i>Summer 2018</i>
Facebook	Software Engineering Intern, Search Ranking	<i>Summer 2017</i>
Google	Software Engineering Intern, Location Segmentation	<i>Summer 2016</i>
Symantec	Machine Learning Intern, Center for Advanced ML	<i>Summer 2015</i>

SELECTED PUBLICATIONS

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- B. Liang, J. Luo, L. Peng, K. Chan, **D. Thaker**, K. A. Kinfu, F. Tian, H. Hassani, R. Vidal. *REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations*. arXiv:2605.12813 *ICML 2026*
 - N. Nguyen, **D. Thaker**, K. Emmanouilidis, T. Ding, R. Vidal. *SSCD: Sparse Semantic Concept Defense Against Semantic Adversarial Attacks to Image Classifiers* *ICLR-W 2026*
 - **D. Thaker**, M. Mostapha, R. Miron, S. Qiu, M. Nadar. *Scale-Cascaded Diffusion Models for Super-Resolution in Medical Imaging*. arXiv:2601.23201 *ISBI 2026 (oral)*
 - B. Liang, L. Peng, J. Luo, **D. Thaker**, K. Chan, R. Vidal. *SECA: Semantically Equivalent & Coherent Attacks for Eliciting LLM Hallucinations*. arXiv:2510.04398 *NeurIPS 2025*
 - **D. Thaker**, A. Goyal, R. Vidal. *Frequency Guided Posterior Sampling for Diffusion-Based Image Restoration*. arXiv:2411.15295 *ICCV 2025*
 - B. Liang, K. Chan, **D. Thaker**, J. Luo, R. Vidal. *KDA: A Knowledge-Distilled Attacker for Generating Diverse Prompts to Jailbreak LLMs*. arXiv:2502.05223 *Preprint 2025*
 - J. Luo, T. Ding, K. Chan, **D. Thaker**, A. Chattopadhyay, C. Callison-Burch, R. Vidal. *PaCE: Parsimonious Concept Engineering for Large Language Models*. arXiv:2406.04331 *NeurIPS 2024*
 - **D. Thaker**, P. Giampouras, R. Vidal. *A Linearly Convergent GAN Inversion-based Algorithm for Reverse Engineering of Deceptions*. arXiv:2306.04756 *Preprint 2023*
 - **D. Thaker***, P. Giampouras*, R. Vidal. *Reverse Engineering ℓ_p Attacks: A Block-Sparse Optimization Approach with Recovery Guarantees*. arXiv:2203.04886 *ICML 2022*

TEACHING

Teaching Assistant, University of Pennsylvania	<i>Fall 2023/Fall 2024</i>
<i>Deep Generative Models (Prof. René Vidal).</i>	
Course Assistant, Columbia University	<i>2019</i>
<i>Deep Learning (Prof. Iddo Drori), Analysis of Algorithms (Prof. Alexandr Andoni).</i>	

HONORS AND AWARDS

ICCV Doctoral Consortium	<i>2025</i>
AWS-AI ASSET Fellow, University of Pennsylvania	<i>2024</i>
Course Assistant Fellowship (full tuition waiver), Columbia University	<i>2019</i>
Turing Scholars Honors, The University of Texas at Austin	<i>2018</i>
University Honors, The University of Texas at Austin	<i>2014–2018</i>

SKILLS

Programming	Python, Bash, C++
ML	PyTorch, TensorFlow, Hugging Face Transformers, Diffusers
Distributed GPU Computing	PyTorch Distributed, CUDA, Triton
Infrastructure	SLURM, AWS